Section 4.3 — Auxiliary Equations with Complex Roots

Active Recall Study Guide

Contents

1 Core Definitions and Theory

Definition: Complex Conjugate Roots of the Auxiliary Equation

Given the homogeneous linear ODE with real constant coefficients:

$$ay'' + by' + cy = 0$$

its *auxiliary (characteristic) equation* is:

$$ar^2 + br + c = 0$$

When $b^2 - 4ac < 0$, the discriminant is negative, so the roots are the complex conjugate pair:

$$r_1 = \alpha + i\beta, \quad r_2 = \alpha - i\beta$$

where

$$\alpha = -\frac{b}{2a}, \qquad \beta = \frac{\sqrt{4ac - b^2}}{2a} > 0$$

Definition: Euler's Formula

For any real θ:

$$e^{i\theta} = \cos\theta + i\sin\theta$$

Derivation sketch (via Maclaurin series): Expand $e^{i\theta}$ using the power series for e^z, observe $i^2 = -1$, and group real/imaginary parts to recover the cosine and sine series.

Consequence:

$$e^{(\alpha+i\beta)t} = e^{\alpha t}(\cos\beta t + i\sin\beta t)$$

Theorem/Lemma: Lemma 2 — Real Solutions from Complex Solutions (p. 167)

Let $z(t) = u(t) + iv(t)$ be a complex-valued solution of $ay'' + by' + cy = 0$ where $a, b, c \in \mathbb{R}$.
Then: both the real part $u(t)$ and imaginary part $v(t)$ are real-valued solutions individually.

Proof: Since $az'' + bz' + cz = 0$, substitute $z = u + iv$:

$$a(u'' + iv'') + b(u' + iv') + c(u + iv) = 0$$

$$\underbrace{(au'' + bu' + cu)}_{\text{real part}=0} + i\underbrace{(av'' + bv' + cv)}_{\text{imag part}=0} = 0$$

A complex number equals zero $\iff$ both its real and imaginary parts are zero. ■

Why this matters: Applying Lemma 2 to $e^{(\alpha+i\beta)t} = e^{\alpha t}\cos\beta t + ie^{\alpha t}\sin\beta t$ gives two independent real solutions: $e^{\alpha t}\cos\beta t$ and $e^{\alpha t}\sin\beta t$.

Theorem/Lemma: General Solution — Complex Conjugate Roots (p. 168)

If the auxiliary equation has complex conjugate roots $\alpha \pm i\beta$ (with $\beta \neq 0$), then a **general real solution** to $ay'' + by' + cy = 0$ is:

$$\boxed{y(t) = c_1 e^{\alpha t}\cos\beta t + c_2 e^{\alpha t}\sin\beta t}$$

where c_1, c_2 are arbitrary real constants.

Linear independence: These two solutions are linearly independent (their Wronskian $W = \beta e^{2\alpha t} \neq 0$ for $\beta \neq 0$), so this truly is a *general* solution.

Definition: Physical Interpretation — Mass-Spring System (p. 169)

The ODE $my'' + by' + ky = 0$ models a damped oscillator where:

- $\alpha = -b/(2m)$: exponential decay rate (negative $\Rightarrow$ damping)
- $\beta = \sqrt{4mk - b^2}/(2m)$: angular oscillation frequency
- $b > \sqrt{4mk}$: overdamping — no oscillation
- $b < \sqrt{4mk}$: underdamping — oscillatory decay
- $b < 0$: negative damping — growing oscillations (energy input)

2 Master Algorithm

Algorithm: Solving $ay'' + by' + cy = 0$ with Complex Roots

Step 1. Write the auxiliary equation: $ar^2 + br + c = 0$.

Step 2. Compute the discriminant: $\Delta = b^2 - 4ac$.
If $\Delta < 0$, roots are complex; proceed. If $\Delta \geq 0$, use Section 4.2 methods.

Step 3. Find roots via quadratic formula:

$$r = \frac{-b \pm \sqrt{\Delta}}{2a} = \alpha \pm i\beta$$

Extract $\alpha = -b/(2a)$ and $\beta = \sqrt{-\Delta}/(2a) > 0$.

Step 4. Write general solution:

$$y(t) = c_1 e^{\alpha t}\cos\beta t + c_2 e^{\alpha t}\sin\beta t$$

Step 5. (IVP only) Apply initial conditions:
Compute $y'(t)$ by product rule, then substitute $t = t_0$ to get a 2×2 linear system for c_1, c_2.

3 Worked Problems

3.1 Problem 5: $w'' + 4w' + 6w = 0$

Solution: Problem 5

Step 1. Auxiliary equation: $r^2 + 4r + 6 = 0$.

Step 2. $\Delta = 16 - 24 = -8 < 0$. Complex roots confirmed.

Step 3.

$$r = \frac{-4 \pm \sqrt{-8}}{2} = \frac{-4 \pm 2i\sqrt{2}}{2} = -2 \pm i\sqrt{2}$$

So $\alpha = -2$, $\beta = \sqrt{2}$.

Step 4. General solution:

$$\boxed{w(t) = c_1 e^{-2t} \cos(\sqrt{2}\, t) + c_2 e^{-2t} \sin(\sqrt{2}\, t)}$$

Physical note: $\alpha < 0$ means decaying (underdamped) oscillations.

3.2 Problem 10: $y'' + 4y' + 8y = 0$

Solution: Problem 10

Step 1. Auxiliary equation: $r^2 + 4r + 8 = 0$.

Step 2. $\Delta = 16 - 32 = -16 < 0$.

Step 3.

$$r = \frac{-4 \pm \sqrt{-16}}{2} = \frac{-4 \pm 4i}{2} = -2 \pm 2i$$

So $\alpha = -2$, $\beta = 2$.

Step 4. General solution:

$$\boxed{y(t) = c_1 e^{-2t} \cos(2t) + c_2 e^{-2t} \sin(2t)}$$

3.3 Problem 16: $y'' + 10y' + 41y = 0$

Solution: Problem 16

Step 1. Auxiliary equation: $r^2 + 10r + 41 = 0$.

Step 2. $\Delta = 100 - 164 = -64 < 0$.

Step 3.

$$r = \frac{-10 \pm \sqrt{-64}}{2} = \frac{-10 \pm 8i}{2} = -5 \pm 4i$$

So $\alpha = -5$, $\beta = 4$.

Step 4. General solution:

$$\boxed{y(t) = c_1 e^{-5t}\cos(4t) + c_2 e^{-5t}\sin(4t)}$$

3.4 Problem 22: IVP $y'' + 2y' + 17y = 0$; $y(0) = 1$, $y'(0) = -1$

Solution: Problem 22

Step 1–3. Auxiliary: $r^2 + 2r + 17 = 0$; $\Delta = 4 - 68 = -64$.

$$r = \frac{-2 \pm 8i}{2} = -1 \pm 4i \implies \alpha = -1,\ \beta = 4$$

Step 4. General solution:

$$y(t) = c_1 e^{-t}\cos(4t) + c_2 e^{-t}\sin(4t)$$

Step 5. Differentiate (product rule applied to each term):

$$y'(t) = e^{-t}\big[(-c_1 + 4c_2)\cos(4t) + (-4c_1 - c_2)\sin(4t)\big]$$

Apply $y(0) = 1$:

$$c_1 \cdot 1 + c_2 \cdot 0 = 1 \implies c_1 = 1$$

Apply $y'(0) = -1$:

$$(-c_1 + 4c_2)\cdot 1 + 0 = -1 \implies -1 + 4c_2 = -1 \implies c_2 = 0$$

Solution:

$$\boxed{y(t) = e^{-t}\cos(4t)}$$

3.5 Problem 27: Third-Order IVP $y''' - 4y'' + 7y' - 6y = 0$; $y(0) = 1$, $y'(0) = 0$, $y''(0) = 0$

Solution: Problem 27

Big Idea: For a 3rd-order ODE, the auxiliary equation is cubic. We must find *three* linearly independent solutions. If the cubic has one real root and one complex conjugate pair, the general solution combines an exponential with the complex-root oscillatory pair.

Step 1. Auxiliary equation: $r^3 - 4r^2 + 7r - 6 = 0$.

Step 2. Test rational roots (by rational root theorem, candidates: $\pm1, \pm2, \pm3, \pm6$):

$$r = 2 : \quad 8 - 16 + 14 - 6 = 0 \checkmark$$

Factor out $(r - 2)$ via synthetic division:

$$r^3 - 4r^2 + 7r - 6 = (r - 2)(r^2 - 2r + 3)$$

Step 3. Solve $r^2 - 2r + 3 = 0$: $\Delta = 4 - 12 = -8$.

$$r = \frac{2 \pm 2i\sqrt{2}}{2} = 1 \pm i\sqrt{2} \implies \alpha = 1,\ \beta = \sqrt{2}$$

Step 4. Three linearly independent solutions: e^{2t}, $e^t \cos(\sqrt{2}\,t)$, $e^t \sin(\sqrt{2}\,t)$.
General solution:

$$y(t) = c_1 e^{2t} + c_2 e^t \cos(\sqrt{2}\,t) + c_3 e^t \sin(\sqrt{2}\,t)$$

Step 5. Compute $y'(t)$ and $y''(t)$, then apply ICs at $t = 0$. Setting up the 3×3 system:
$y(0) = 1$:

$$c_1 + c_2 = 1$$

$y'(0) = 0$: Differentiate and evaluate. The derivative at $t = 0$ gives:

$$2c_1 + (c_2 + \sqrt{2}\,c_3) = 0$$

$y''(0) = 0$: Second derivative at $t = 0$ gives:

$$4c_1 + (c_2 - 2c_3 + 2\sqrt{2}\,c_3 - 2c_2) \text{ [collect carefully]}$$

Let us be explicit. For $e^t \cos(\sqrt{2}\,t)$: first derivative is $e^t \cos(\sqrt{2}\,t) - \sqrt{2}\,e^t \sin(\sqrt{2}\,t)$, second is $e^t(\cos(\sqrt{2}\,t) - 2\sqrt{2}\sin(\sqrt{2}\,t) - 2\cos(\sqrt{2}\,t)) = e^t(-\cos(\sqrt{2}\,t) - 2\sqrt{2}\sin(\sqrt{2}\,t))$. For $e^t \sin(\sqrt{2}\,t)$: second derivative is $e^t(-\sin(\sqrt{2}\,t) + 2\sqrt{2}\cos(\sqrt{2}\,t))$.
At $t = 0$:

$$y''(0) : \quad 4c_1 + (-1)c_2 + (2\sqrt{2})c_3 = 0$$

The 3×3 linear system:

$$\begin{pmatrix} 1 & 1 & 0 \\ 2 & 1 & \sqrt{2} \\ 4 & -1 & 2\sqrt{2} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

From row 1: $c_1 = 1 - c_2$. Substitute into row 2: $2(1 - c_2) + c_2 + \sqrt{2}c_3 = 0 \Rightarrow 2 - c_2 + \sqrt{2}c_3 = 0$. Substitute into row 3: $4(1 - c_2) - c_2 + 2\sqrt{2}c_3 = 0 \Rightarrow 4 - 5c_2 + 2\sqrt{2}c_3 = 0$. From the two equations in c_2, c_3:

$$-c_2 + \sqrt{2}c_3 = -2 \quad \text{(A)}$$

$$-5c_2 + 2\sqrt{2}c_3 = -4 \quad \text{(B)}$$

$(B) - 2(A)$: $-3c_2 = 0 \Rightarrow c_2 = 0$.
From (A): $\sqrt{2}c_3 = -2 \Rightarrow c_3 = -\sqrt{2}$.
Then $c_1 = 1$.

Final solution:

$$\boxed{y(t) = e^{2t} - \sqrt{2}\, e^t \sin(\sqrt{2}\, t)}$$

3.6 Problem 30: Verify differentiation formula $\frac{d}{dt}e^{(\alpha+i\beta)t} = (\alpha + i\beta)e^{(\alpha+i\beta)t}$

Solution: Problem 30

Big Idea: The familiar rule $\frac{d}{dt}e^{rt} = re^{rt}$ must be *verified* for complex r because we cannot assume it holds without proof. We use the real representation from Euler's formula instead.

Given representation (equation (6)):

$$e^{(\alpha+i\beta)t} = e^{\alpha t}\cos\beta t + i\,e^{\alpha t}\sin\beta t$$

Differentiate both sides with respect to t (real and imaginary parts separately):

$$\frac{d}{dt}e^{(\alpha+i\beta)t} = \frac{d}{dt}\left[e^{\alpha t}\cos\beta t\right] + i\,\frac{d}{dt}\left[e^{\alpha t}\sin\beta t\right]$$

By product rule:

$$= e^{\alpha t}(\alpha\cos\beta t - \beta\sin\beta t) + i\,e^{\alpha t}(\alpha\sin\beta t + \beta\cos\beta t)$$

Group real and imaginary:

$$= \alpha e^{\alpha t}(\cos\beta t + i\sin\beta t) + \beta e^{\alpha t}(-\sin\beta t + i\cos\beta t)$$

Note that $-\sin\beta t + i\cos\beta t = i(\cos\beta t + i\sin\beta t)$ (multiply $\cos\beta t + i\sin\beta t$ by i):

$$= \alpha e^{(\alpha+i\beta)t} + i\beta e^{(\alpha+i\beta)t} = (\alpha+i\beta)e^{(\alpha+i\beta)t} \qquad \blacksquare$$

3.7 Problem 35: Swinging Door — Determine values of b for no sustained oscillation

Solution: Problem 35

Big Idea: The swinging door ODE is structurally identical to a damped mass-spring. The question asks: for which damping values b does the door *not* oscillate (i.e., returns to closed without swinging back and forth)?

The IVP:

$$I\theta'' + b\theta' + k\theta = 0, \quad I > 0,\ b > 0,\ k > 0$$

The auxiliary equation is $Ir^2 + br + k = 0$, with discriminant:

$$\Delta = b^2 - 4Ik$$

Case analysis (logical disjunction on Δ):

$\Delta < 0$ ($b^2 < 4Ik$): roots are complex $\Rightarrow$ general solution is $e^{\alpha t}(c_1\cos\beta t + c_2\sin\beta t)$ with $\alpha = -b/(2I) < 0$. The door **oscillates** (underdamped).

$\Delta = 0$ ($b^2 = 4Ik$): repeated real root $r = -b/(2I) < 0$. Solution $e^{rt}(c_1 + c_2 t) \to 0$ monotonically. Door **does not oscillate** (critically damped).

$\Delta > 0$ ($b^2 > 4Ik$): two distinct real negative roots. Solution is sum of decaying exponentials. Door **does not oscillate** (overdamped).

Conclusion: The door will *not* continually swing back and forth if and only if:

$$\boxed{b \geq 2\sqrt{Ik}}$$

(critical damping or overdamping). For $0 < b < 2\sqrt{Ik}$, the door oscillates.

4 Practice Problems (for Problems 27 and 35)

4.1 Practice for Problem 27 (Third-Order IVP)

Practice Problem: P1 — Third-Order IVP with Mixed Roots

Solve the initial value problem:

$$y''' - 3y'' + 4y' - 2y = 0; \quad y(0) = 0,\ y'(0) = 1,\ y''(0) = 0$$

Hint: $r = 1$ is a root of the auxiliary equation. Factor it out and solve the remaining quadratic for a complex conjugate pair.

Practice Problem: P2 — Third-Order IVP with Different Real Root

Solve the initial value problem:

$$y''' - 2y'' + 5y' - 4y = 0; \quad y(0) = 1,\ y'(0) = 1,\ y''(0) = 3$$

Hint: $r = 1$ is a root. The remaining factor gives roots $\alpha \pm i\beta$. Write the general solution as $c_1 e^t + e^t(c_2 \cos \beta t + c_3 \sin \beta t)$, then solve the 3×3 system.

4.2 Practice for Problem 35 (Damping Threshold)

Practice Problem: P3 — RLC Circuit Oscillation Threshold

An RLC circuit satisfies $Lq'' + Rq' + \frac{1}{C}q = 0$ with $L = 4\,\mathrm{H}$, $C = \frac{1}{9}\,\mathrm{F}$, and $R > 0$.

For what values of R does the charge $q(t)$ oscillate? For what values does it not oscillate?

Hint: Identify the structural analogy: $L \leftrightarrow I$, $R \leftrightarrow b$, $1/C \leftrightarrow k$. Apply the same discriminant threshold.

Practice Problem: P4 — Threshold with Symbolic Parameters

A door has moment of inertia $I = 2\,\mathrm{kg\,m^2}$ and spring constant $k = 8\,\mathrm{N\,m/rad}$.

(a) Find the critical damping value b^*.

(b) If $b = 4\,\mathrm{N\,m\,s/rad}$, classify the motion and find the general solution form.

(c) If $b = 12\,\mathrm{N\,m\,s/rad}$, classify the motion and find the general solution form.

5 Key Propositional Logic Summary

Decision Tree: Second-Order Constant-Coefficient ODE

Let P: "$\Delta = b^2 - 4ac < 0$" Q: "$\Delta = 0$" R: "$\Delta > 0$"

$P \Rightarrow$ complex roots $\alpha \pm i\beta \Rightarrow$ oscillatory solution $e^{\alpha t}(c_1 \cos \beta t + c_2 \sin \beta t)$

$Q \Rightarrow$ repeated real root $r = -b/(2a) \Rightarrow$ solution $(c_1 + c_2 t)e^{rt}$

$R \Rightarrow$ two distinct real roots $r_1, r_2 \Rightarrow$ solution $c_1 e^{r_1 t} + c_2 e^{r_2 t}$

$\alpha < 0$: decaying oscillation $\alpha = 0$: pure sinusoidal $\alpha > 0$: growing oscillation